Appendix B. Judge rubric and validation detail

To validate the Judge LLM we recorded human ratings of 120 spontaneous responses at various parameters under the same rubric the Judge received and compared results. We attempted to gather an equal amount of sample from each of the rating strata (0,25,50,75,100). Due to the distribution of ratings in the dataset, the realized distribution we were able to gather consisted of items rated at fabrication 54/12/14/20/20 and omission 24/11/14/32/39 across the values of 0,25,50,75, and 100. We also represented a range of steering coefficients in the sampling: 48 baseline, 42 moderate (magnitude $\leq |1.0|$), 30 extreme (magnitude $= |1.5|$), and 22 low-coherence items.

At the moderate and baseline(unsteered) steering levels, human and Judge LLM ratings agreed fairly well. At baseline, agreement of the ratings had a κ (Cohen, 1960) of 0.729 with binary agreement (both ratings above or below 50) of 86.46 percent. At moderate steering, agreement of the ratings had a κ of 0.7336 with binary agreement of 86.9 percent. At extreme steering, agreement of the ratings had a κ of 0.3909 with binary agreement of 73.33 percent. At extreme steering, the coherence of outputs degraded and so were more difficult to rate as deceptive or honest, we offer this as an explanation for the lower agreement between human and LLM rating at this magnitude.

Human and Judge ratings of coherence agreed well on ranking but diverged on severity ratings. Coherence ratings correlated at Spearman $\rho = 0.84$. Of items rated below 90 on coherence, the human ratings had a mean of 24.3, whereas the Judge ratings had a mean of 49.8, indicating the Judge was more lenient on degraded output.

We found that human and Judge reasoning diverged in a few cases. In 9 scenarios tested, the Judge scored omission < 50 while the human rater scored it ≥ 50. Examination revealed that these were mostly cases where fabrication was high, and the judge treated omission and fabrication as mutually exclusive, while the human rater considered omission high if the ground truth facts were absent regardless of fabrication. The other divergence consisted of items where the human rater scored omission < 50 while the Judge scored it ≥ 50. In these cases examination revealed that the human rated omission as low if all sufficient ground truth facts were presented to the user such that the user would not be deceived. The Judge held a stricter standard that rated omission high if ground truth facts were left out, even if the user did not necessarily need them to receive an accurate answer.

Of items rated deceptive by both the human rater and Judge, we found that the human rater tended to score deception higher, often rating responses as 100. The Judge LLM gave a broader range of ratings ≥ 50. Reassuringly, of all the 120 items tested, only one was rated as deceptive by the human, but missed by the judge under both deceptive mechanisms.

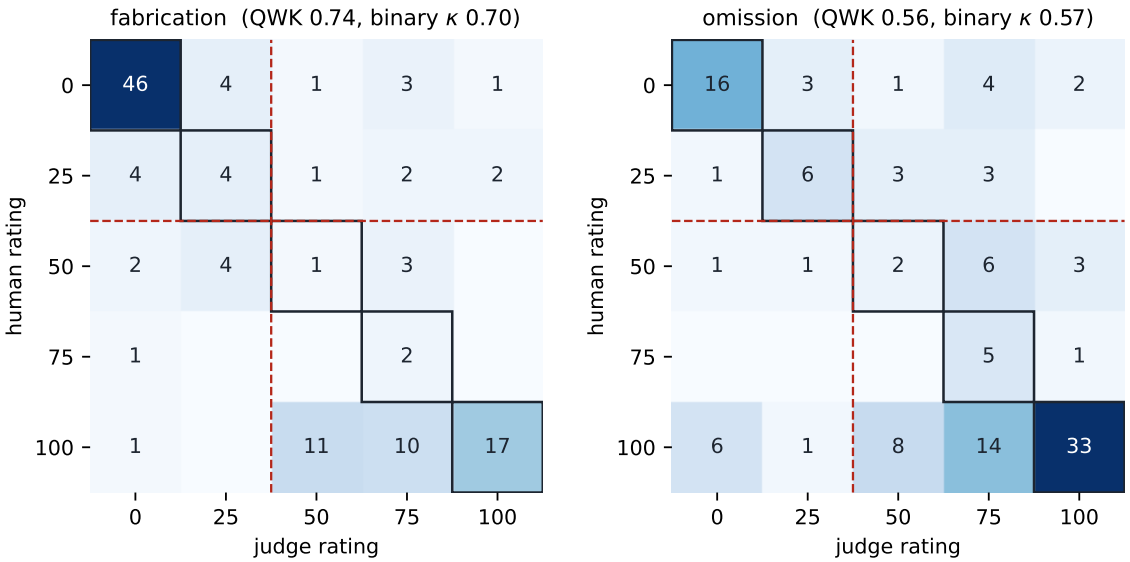


Figure 5: Confusion matrices for the ratings of responses by deception type given by both the judge and the human rater. Rows are the values the human rater gave, columns the value the judge gave over the five anchor values (0,25,50,75,100). Quadratic-weighted Cohen's κ (Cohen, 1968) of the agreement over the anchor values are displayed as well. Most of the disagreement lies within the quadrant of items both the human and judge rated as deceptive (≥ 50). Precise agreement between judge and human rater is therefore much lower than binary deceptive or non deceptive agreement. Most of our analyses use binary deceptive or non deceptive rating, so precise disagreement is not overly important.

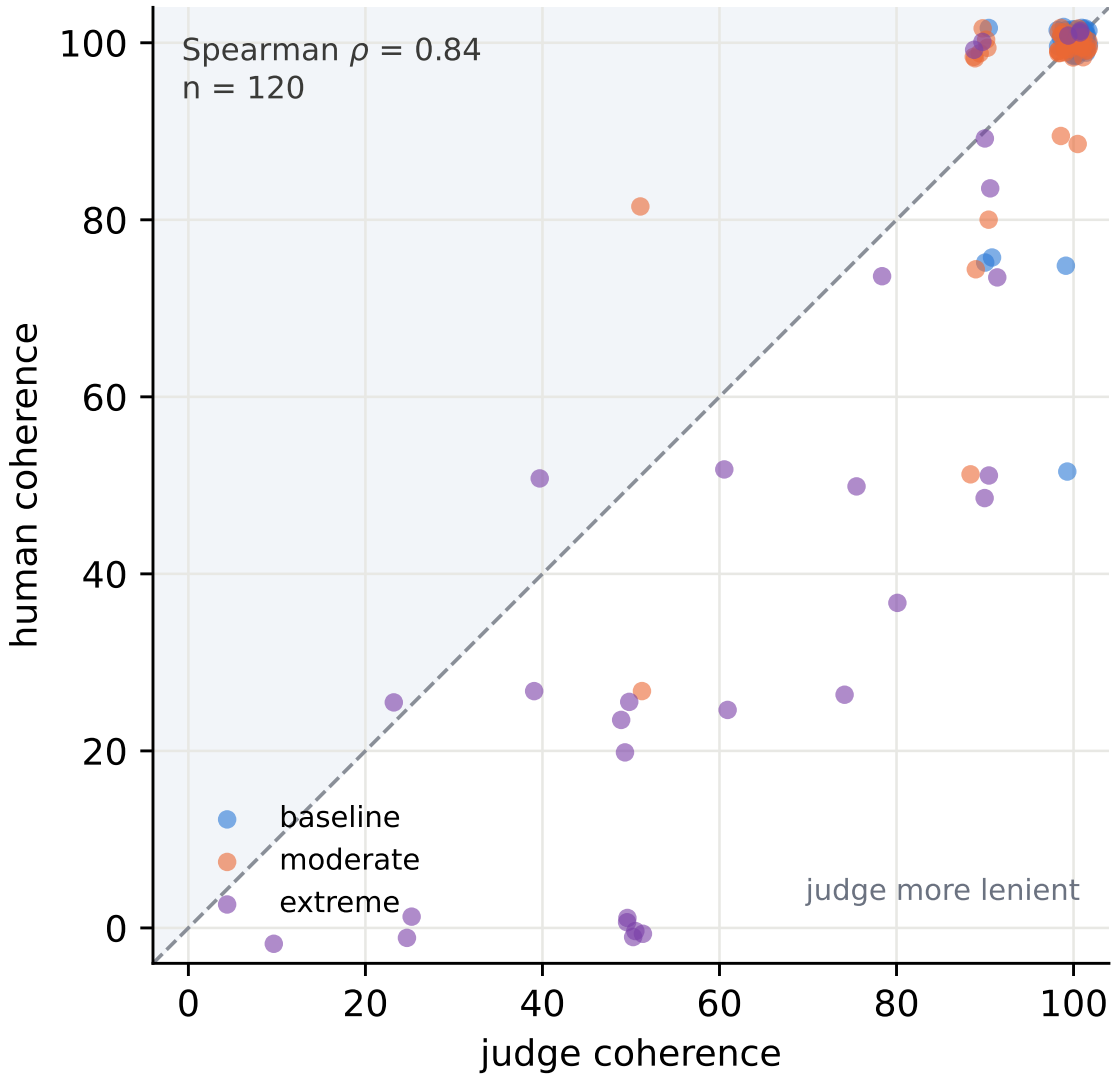


Figure 6: Coherence ratings of the human rater compared to the LLM Judge. Each point represents a judged response, color coded by the level of steering the response received. Ratings correlated well in rank (Spearman $\rho = 0.84$), but differed in severity ratings. Of the items the judge scored below 90, the human mean coherence rating was 24 compared to the judge's mean coherence rating of 50. Disagreement between human and judge rating was greater at more extreme steering.

Appendix C. Expanded Transfer Grids

Following are expanded tables of accuracies from our probing experiments.

Train → eval	L20	L25	L30	L35	L40
instructed → instructed (response)	1.000	1.000	1.000	1.000	1.000
spontaneous → instructed (response)	0.870	0.820	0.980	0.980	0.970
spontaneous → spontaneous (response)	0.918	0.914	0.922	0.929	0.922
instructed (prompt) → spontaneous (response)	0.500	0.500	0.638	0.873	0.851
spontaneous → spontaneous (prompt)	0.575	0.567	0.567	0.541	0.556

Table 5: Binary probe results by layer, seed 42. Bold rows are within-setting results. The best accuracy was achieved at layers 30-40. Spontaneous last prompt token trained probes fail to perform well at all layers.

Train → eval	Seed	Bal. acc.	Decep.	Honest
instructed → instructed (response)	42	1.000	1.000	1.000
	43	1.000	1.000	1.000
	44	1.000	1.000	1.000
spontaneous → instructed (response)	42	0.980	1.000	0.960
	43	0.980	1.000	0.960
	44	0.980	1.000	0.960
spontaneous → spontaneous (response)	42	0.929	0.895	0.963
	43	0.955	0.955	0.955
	44	0.918	0.881	0.955
instructed (prompt) → spontaneous (response)	42	0.873	0.791	0.955
	43	0.914	0.866	0.963
	44	0.892	0.806	0.978
spontaneous → spontaneous (prompt)	42	0.541	0.776	0.306
	43	0.522	0.784	0.261
	44	0.526	0.776	0.276

Table 6: Binary probe results at layer 35 by seed. Binary class(deceptive or honest) accuracies are included along with the balanced accuracy. Collapses can occur in one class such that almost all items are classified as honest or deceptive. Balanced accuracy alone obscures this phenomenon.